

EL GHODBANE Natej

Seeking a Data / AI Apprenticeship – Master's Degree (M2 SIA)

+33 7 46 65 10 15 – ghodbanenatej@gmail.com – LinkedIn – GitHub – Portfolio – Paris / Île-de-France

Master's student in Intelligent Systems and Applications at Université Gustave Eiffel, seeking a **12-month Data / AI apprenticeship**, available immediately, on a **1 week at university / 1 week in the company** schedule. Experience in Data Science, MLOps and Generative AI, notably with Python, SQL, LLMs, LangGraph and RAG. Autonomous, curious and comfortable solving complex technical problems.



Skills

Programming Languages: Python, SQL, R, Java, JavaScript, C++, C, PHP

Data Science & ML: Pandas, NumPy, Scikit-Learn, TensorFlow, PyTorch, Matplotlib

Generative AI & LLMs: LLMs, LangGraph, RAG, MCP, Function Calling, Prompt Engineering, Context Engineering

Databases / MLOps: MySQL, PostgreSQL, MongoDB, MongoDB Atlas, MLflow, Docker, Jenkins, Elasticsearch, Kibana, Logstash

Frameworks / BI / Tools: FastAPI, Flask, Next.js, Symfony, Streamlit, Power BI, DAX, Git, GitHub, Talend, Excel, JavaFX, FlutterFlow

Professional Experience

Kernel Solutions & Innovation – AI Engineer Intern

June 2026 – August 2026

- Designed and developed end-to-end a 100% on-premise AI coding assistant combining local LLMs, RAG and AI agents, adopted by **4 internal developers** during the testing phase.
- Implemented traceability ensuring **100% auditability** of generated responses and deployed the stack with Docker, reducing setup time from several hours to **under 15 minutes**.
- Eliminated all data transfers to third-party services using Ollama and Qdrant, ensuring **100% data sovereignty**.

Tools: Python, FastAPI, Ollama, LangGraph, Qdrant, Tree-sitter, LLMs, RAG, Docker, Git

BeeCoders – Data Science Intern

June 2025 – July 2025

- Collected, cleaned and structured a dataset of more than **5,000 records**, improving data quality for training **3 ML models**.
- Developed a recommendation system based on customer data, increasing recommendation relevance by **20%**.
- Implemented an LLM- and RAG-based chatbot, improving response accuracy by **30%** compared with the initial version.

Tools: Python, Pandas, NumPy, Scikit-Learn, Flask, Talend, Excel, LLMs, RAG

BH Bank – Data Analyst Intern

August 2023

- Designed and implemented a Power BI dashboard to visualize and monitor key indicators derived from banking data.
- Participated in consistency checks for several dozen financial reports.
- Gained insight into business processes across **3 departments**: front office, back office and compliance.

Tools: Power BI, Excel, Python

Projects

Firmaa – Intelligent Agriculture Platform

2026

- AI-driven agricultural decision-support platform using autonomous agents and real-time data; disease detection and irrigation optimization models achieving **>90% accuracy**.

Tools: LangGraph, FastAPI, Next.js, MongoDB Atlas, TensorFlow, LLMs

Breast Cancer Detection & Risk Stratification System

2026

- Benign/Malignant classification and risk stratification system (30 features); FastAPI backend with optimized logistic regression achieving **98.83% accuracy**.

Tools: Python, Scikit-Learn, FastAPI, Next.js, TypeScript, Tailwind CSS

MLOps Churn Prediction Platform

2026

- Churn prediction platform using XGBoost, FastAPI, Docker, Jenkins and MLflow; **AUC-ROC 0.92**, experiment tracking and monitoring with ELK.

Tools: Python, XGBoost, FastAPI, Docker, Jenkins, MLflow, ELK

Education

Université Gustave Eiffel – Master's Degree in Intelligent Systems and Applications (SIA) 2026 – 2027

ESPRIT – Computer Engineering Degree, Data Science Specialization

2022 – Present

Certifications & Languages

Certifications: Fundamentals of Deep Learning – NVIDIA; Application of AI for Anomaly Detection – NVIDIA

Languages: French (C1); English (C1)